NAME: TANKISO MASOEBE

COMPUTER SECURITY WEEK 11

1. Home Network Security Audit Report

1. Introduction

The purpose of this audit is to evaluate the security of my home Wi-Fi network, identify possible vulnerabilities, and recommend ways to improve its security.

2. Security Audit Findings

During the audit, the following network security details were considered:

- **SSID Name:** Home Wi-Fi
- **Security Protocol:** WPA2-PSK
- **WPS Status:** Enabled
- **Router Firmware Version:** Needs to be checked in the router settings.

The network may have several security vulnerabilities. Using WPA2 with a weak password can make the network vulnerable to unauthorized access. Leaving WPS enabled may increase the risk of attackers gaining access to the Wi-Fi network. Outdated router firmware can also contain security weaknesses that attackers may exploit.

3. Steps to Improve Network Security

To improve the security of my home network, I would take the following steps:

1. Upgrade the security protocol to WPA3-Personal if the router supports it.
2. Change the Wi-Fi password to a strong and unique password containing letters, numbers, and symbols.
3. Disable WPS to reduce the risk of unauthorized access.
4. Update the router firmware to the latest version provided by the manufacturer.
5. Change the default router administrator password to prevent unauthorized changes to network settings.
6. Regularly check connected devices and remove any unknown devices.
7. Create a separate guest network for visitors to protect personal devices.

4. Conclusion

The home network security audit shows that proper security settings are important for protecting personal information and preventing unauthorized access. By using strong passwords, disabling WPS, updating firmware, and monitoring connected devices, I can improve the security and reliability of my home Wi-Fi network.

2. Research: The Future of Wireless Security – 5G Security

Introduction

5G is the fifth generation of mobile network technology. It provides faster internet speeds, lower latency, and the ability to connect many devices simultaneously. As technology continues to develop, 5G security is becoming increasingly important for protecting personal information and preventing cyberattacks.

Primary Use Cases of 5G Technology

5G technology is used in smart cities, healthcare, industrial automation, autonomous vehicles, and mobile communication. For example, hospitals can use connected medical devices to monitor patients remotely. Industries can use 5G to control machines and robots, while smart cities can use it to manage traffic lights, security cameras, and other connected systems. Its high speed and low latency make communication between devices faster and more efficient.

Unique Security Challenges

Despite its advantages, 5G introduces several security challenges. One major challenge is the increased number of connected devices, which creates more opportunities for cyberattacks. Attackers may target vulnerable Internet of Things (IoT) devices to gain unauthorized access to networks. Another challenge is the use of virtualized network functions and cloud-based infrastructure, which can introduce software vulnerabilities if not properly secured. Furthermore, distributed network equipment and insecure configurations may expose sensitive information to attackers. Denial-of-service attacks can also disrupt communication services.

Recommended Security Measures

Several measures can improve 5G security. First, network operators should use strong authentication and encryption to protect communication and prevent unauthorized access. Second, software and network equipment should be updated regularly to fix known security vulnerabilities. Third, organizations should use firewalls, intrusion detection systems, and continuous network monitoring to identify suspicious activities. Network segmentation should also be implemented to separate critical systems from less secure devices. Finally, organizations should follow established security standards and conduct regular security audits.

Conclusion

5G technology offers significant benefits in communication, healthcare, industry, and smart cities. However, its growing use creates new cybersecurity risks. Implementing strong authentication, encryption, regular updates, and network monitoring will help protect 5G networks and ensure secure and reliable communication in the future.